

Task 5: Dynamic Quoting Under Inventory Pressure

The quoting function $Q(I, \sigma, \alpha, \eta)$ returns half-spreads (δ_b, δ_a) measured relative to mid, so the constraint $c_{\min} \cdot \sigma \leq \delta \leq \delta_{\max}$ (with $c_{\min}=0.5$ and $\delta_{\max}=50\text{bps}$) is dimensionless. Quoted prices are $b=M-\delta_b$, $a=M+\delta_a$. A fill at half-spread $\delta = x\sigma$ earns, per unit volume, a trade PnL of $e+x\sigma$, where $e=\text{side} \cdot (M_{\text{close}} - M_0)/M_0$ is the realised mid-edge and $x\sigma$ is the captured spread. With fill probability $\lambda e^{-\gamma x}$, the expected reward per arrival is $\lambda e^{-\gamma x} (E[e] + x\sigma)$, whose maximizer is the objective-optimal half spread $x^*=1/\gamma - \beta(\alpha)$ (volume cancels). We quote a symmetric base $\delta=\sigma x^*$, add an inventory skew and clip to $[c_{\min} \sigma, \delta_{\max}]$. The function keeps light internal state across calls so it can react to the undisclosed regime shifts using the observable signal alone.

Parameterization and intuition:

The design uses one observable mechanism per kind of regime shift, so no hidden parameter (λ, γ or ϕ) is ever needed:

- Fill regime (λ, γ): γ is estimated online from observed fills (the inventory change between two calls reveals whether the last quote filled, and the slope of the log fill-rate on $x=\delta/\sigma$ is $-\gamma$. λ cancels in the slope, so the estimate is λ invariant. It sets the $1/\gamma$ term of x^*
- Adversity: $\beta(\alpha)=E[e/\sigma|\alpha]$ is calibrated from the data ($\beta \sim 5.84 - 12.95\alpha$). Positive for benign flow, so we quote tighter than $1/\gamma$ to capture profitable volume, and negative for toxic flow, so we quote wider to avoid adverse fills. The popular $x^*=1/\gamma$ is only the $E[e]=0$ special case. This maximises the captured spread, not realised PnL. An optional realised PnL feedback (SPSA) corrects residual beta error when a desk can see its 30 second PnL.
- Penalty (ϕ): An Avellaneda-Stoikov tanh inventory skew with end-of-day urgency $(1 + k_{\eta}\eta)$ flattens inventory toward zero by the close without knowing ϕ . This is an asymmetric-risk choice, since flattening is cheap but carrying inventory under a large unknown ϕ is quadratically costly.
- Volatility (σ): The base is σ -proportional with a shock guard that widens when σ exceeds its EWMA, so the quoted δ^* scales with volatility automatically.

Validation methodology and results:

We compare the objective target against the fill-model target ($x^* = 1/\gamma$) and a non-adaptive static quote, on a stationary (λ, γ, ϕ) grid and on a stream with three discontinuous regimes (easy $\rightarrow$ hard + toxic + high- $\phi \rightarrow$ medium). On the stationary grid the objective target scores 229 vs 189 (fill-model) vs 180 (static). On the regime-shift stream it reaches PnL 95.5k and Score 51.3, beating the fill-model design by +8% PnL and the static quote by +19%, with zero drawdown. The gain is largest in the benign easy-fill regime, where the $1/\gamma$ target over-widens and sheds profitable volume. The one honest limit is that non-directional toxicity which α does not flag, arriving without PnL feedback, is outside Q's observability; it is defended through α, β , and inventory control rather than claimed as detected.



